The basket may be one-sided (e.g., all buys or all sells) or two-sided (e.g., both buys and sells). Traders are then tasked with determining the most appropriate way to transact the order over a specified period of time. This is accomplished by balancing the trade-off between cost and risk based on a user specified level of risk aversion. Mathematically, this is stated as follows:

$$Min\ Cost(x_k) + \lambda \cdot Risk(x_k)$$

Where λ represents trader specified level of risk aversion, and x_k is used to denote the discrete trade schedule representing exactly how the shares are to be transacted in each period for each stock.

Variables

$X = shares\ to\ trade$

$Y_t = shares\ executed\ at\ time\ t$

$$Side(i) = \begin{cases} +1 & if\ buy\ order \\ -1 & if\ sell\ order \end{cases}$$

$ADV = average\ daily\ volume$

$\sigma = annualized\ volatility$

$C = covariance\ matrix,\ scaled\ for\ the\ length\ of\ the\ trading\ period\ in\ (\$/Share)^2$

$x_{ij} = shares\ of\ stock\ i\ to\ trade\ in\ period\ t$

$r_{ij} = residual\ shares\ of\ stock\ i\ at\ beginning\ of\ period\ t$

$$r_{ij} = \sum_{k=j}^{n} x_{ik}$$

$v_{it} = volume\ for\ stock\ i\ in\ period\ t$

$P_{i0} = arrival\ price$

$P_{it} = market\ price\ in\ period\ t$

$\overline{P}_{it} = average\ execution\ price\ at\ time\ t$

$m = number\ of\ stocks\ in\ the\ portfolio$

$n = number\ of\ trading\ periods\ during\ the\ horizon$

$d = number\ of\ trading\ periods\ per\ day$